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Speech comprehension relies on both temporal structure and contextual prediction. Yet, it is poorly understood how these mechanisms coordinate to assure stable comprehension even under rapid replacement of units unfolding in time. Oscillatory accounts propose that this may be the role of brainwaves, with flexible theta-band oscillations (~4-8 Hz) supporting temporal parsing of speech, while beta-band oscillations (~12-30 Hz) are implicated in maintaining and updating contextual predictions. In the present study, we use time-compression to degrade temporal structure while preserving linguistic content and investigate how information delivery rate, temporal regularity, and linguistic structure jointly shape speech comprehension. Across two behavioral 're-packaging' experiments (N = 110 neurotypical adults), x3 time-compressed sentences are combined with periodic insertions of silences between the speech units to manipulate (i) delivery rate of sound packages, (ii) rhythmic regularity of such delivery, and (iii) boundary alignment (syllable-based vs. strict duration time-based segmentation). In addition, we quantify contextual predictability using word-level entropy to assess the contribution of top-down constraints. Results reveal (i) a non-linear effect of delivery rate: comprehension peaks at rates exceeding the canonical theta range (~8–13 Hz) and declines at slower or faster pacing ($p < .001$ for both Exp 1 and 2). (ii) Temporal regularity benefits word recognition only when boundaries coincide with syllabic units, indicating that periodic pacing alone is insufficient for effective parsing. (iii) Syllable-based alignment of packages significantly improves word recognition relative to time-based segmentation ($p < .001$), with strongest benefits outside the optimal rate range (e.g., 10.8 Hz: $p < .001$). Crucially, contextual predictability (word-level entropy) facilitates comprehension primarily when temporal cues are least supportive: outside the optimal theta range under syllable-aligned segmentation (three-way interaction Exp.1: $p = .048$; Exp.2: $p = .031$). Theta mediation is efficient for comprehension when the speech is delivered within its optimal regime, while fades outside this regime, allowing the active role of beta-mediated predictive processes. Together, these results suggest that contextual prediction is continuously active but behaviorally visible only when temporal scaffolding is insufficient and syllabic structure is preserved.

D88 - Speech Processing in the Wild: EEG research goes to school

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Understanding speech processing in dynamic and noisy real-life environments, is a core translational challenge of cognitive neuroscience, which is not well captured in laboratory experiments. To bridge this gap, instead of bringing participants to the lab, here we brought a mobile-lab to a real-life environment where natural speech-processing and attention to speech are especially challenging: a typical high-school classroom. We recorded neural activity (EEG), gaze and body motions from N=100 students, who viewed educational videos in their organic classroom, interspersed with occasional

ecological disturbances consisting of other students talking. First, we verified that reliable neural speech-tracking responses could be obtained in this real-life mobile-EEG setting, and compared results to a parallel lab-based study using identical material. Next, we tested how learning-outcomes, neural speech tracking, gaze and posture were affected by the presence of background speech disturbances, and characterized the neural response to the disturbances themselves. We found that performance and neural speech tracking were negatively affected by background speech, however analysis of gaze and posture suggest investment of effort to maintain attention to the lecture despite the disturbances. We also found significant effects of time-on-task, with all measures associated with attention and speech processing deteriorating over the course of the lesson. Exploratory analysis further reveal interesting individual differences in both the effect that background disturbances have on speech processing and the trajectory of changes over time. This work emphasizes the feasibility and significance of conducting field-based neuroscientific research and demonstrate the importance of considering brain-body-behavior interactions in real-life speech processing.

D89 Sandbox Series - Anatomical auditory–motor white-matter connectivity underlies individual differences in cortical speech tracking

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Auditory–motor connections are crucial for speech production and have been increasingly implicated in speech comprehension, yet whether variation in their structural integrity accounts for interindividual differences in neural speech tracking remains unclear. To address this question, we combined MEG-based neural speech tracking with diffusion-weighted imaging from a large open-access dataset of neurotypical individuals. Neural speech tracking was quantified using multivariate Gaussian copula mutual information. MEG time series were extracted for several auditory and speech motor regions of interest. Tracking in the delta band (1–3 Hz) was strongest in the ventral motor cortex and showed greater interindividual variability than in classic language regions, including the superior temporal gyrus and inferior frontal gyrus. Strikingly, this variability was predicted by the total volume of the left arcuate fasciculus, a major auditory–motor white-matter pathway. These findings provide initial evidence that auditory–motor connectivity underlies interindividual variability in neural speech tracking within motor regions.

D90 - Daytime nap consolidates learning of Cantonese tones in non-tonal language speakers: sleep-EEG evidence.

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Prior research shows that post-training sleep facilitates the consolidation of newly learned speech sounds, such as consonants. Lexical tones, which use pitch variations to signal word identity, are a useful case for examining sleep-mediated speech learning because they are challenging for adult second-

language learners. A daytime nap study of (Mandarin-speaking) adults who speak tonal languages found neither a nap-specific behavioral benefit nor sleep-EEG support: both the nap and wake groups showed offline consolidation, and sleep-EEG parameters, spindle density, and slow oscillations (SOs) did not predict the nap group's performance. Recent work examining (English-speaking) adults who don't speak tonal languages, however, found that nap facilitated higher-level Mandarin tone categorization relative to wakeful rest, but this nap-specific benefit was not associated with expected sleep parameters, such as NREM sleep. While nap-mediated consolidation of Cantonese tones would likely occur in non-tonal language speakers, its association with sleep-EEG parameters needs further investigation. This study addresses this among the Hong Kong ethnic-minority population from South/Southeast Asia, who speak non-tonal languages and face persistent challenges acquiring spoken Cantonese. Seventy-eight participants with limited spoken Cantonese exposure were randomly assigned to either a nap group or a non-nap group ($n = 39$ for each). They were trained to identify Cantonese words contrasting in contour and level tones (240 trials per contrast, order counterbalanced). After training, participants completed tone identification test 1 with trained stimuli (ID1; 80 trials per contrast) as a pre-test. They then completed a 90-min manipulation: the nap group napped with an EEG recording, whereas the non-nap group remained awake for non-verbal activities. Following this manipulation, participants completed an identical identification test 2 (ID2) as a post-test, with changes in ID2-ID1 indexing consolidation effects. Spindles were identified across the ten scalp electrodes from nap participants' NREM (i.e., N2 and N3) sleep, and SOs were identified in frontal electrodes from NREM-N3 sleep using Yet Another Spindle Algorithm (YASA). Mixed-effects models were fitted to the accuracy of ID1 and ID2 for the nap and non-nap groups. The model revealed a significant Group $\times$ Session interaction ($z = -2.18$, $p = .029$). Post-hoc analyses showed that the nap group improved from ID1 to ID2 ($z = -2.80$, $p = .005$), whereas the non-nap group did not. The analyses of the sleep-EEG data showed that nap participants' spindle density ($z = 3.73$, $p < .001$) and SOs density ($z = 3.01$, $p = .002$) positively predicted consolidation, with better nappers showing greater improvement. The results provide behavioral and sleep-EEG evidence that a daytime nap supports Cantonese tone consolidation in non-tonal language speakers. Specifically, only the nap group showed an ID1-to-ID2 improvement, suggesting nap-mediated offline consolidation of newly learned tone categories. This effect was further supported by sleep-EEG evidence, as higher spindle and SOs density predicted greater ID1-to-ID2 gains in the nap group. Together, these findings suggest that sleep-related neurophysiology contributes to the consolidation of newly learned Cantonese tones, highlighting a key role of sleep-dependent memory consolidation in supporting lexical tone learning among non-tonal language speakers.

D91 - Chills Beyond Music: How Speech Elicits Skin Conductance Responses

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Chills are brief, often pleasurable bodily sensations that are commonly associated with music and are usually accompanied by skin conductance responses, which are regarded as a neurophysiological marker of aesthetic processing. This adds a measurable physiological component to the often behaviorally assessed phenomenon. While music-elicited chills have been studied thoroughly, much less is known about whether spoken language can produce comparable responses. Several accounts propose shared evolutionary, affective, and neurocognitive mechanisms for music and speech, including the musical protolanguage theory and the musi-linguistic continuum. Building on the concept of phonetic chills, the present study examined whether non-melodic speech could elicit physiological chill responses, and whether such responses align with behavioral reports of chills. Seventy-eight participants ($N = 78$) listened to 16 auditory stimuli and provided continuous self-reports and retrospective chill ratings. The experiment took place under controlled laboratory conditions. Electrodermal activity was recorded from electrodes placed on the distal phalanges of the index and middle fingers of the left hand. Recordings were obtained using a g.Hamp biosignal amplifier at 256 Hz, alongside pulse, plethysmogram, and SpO₂ measures. Preprocessing and feature extraction were conducted in Python using open-source tools including NeuroKit2, EDA-Explorer, NumPy, SciPy, and pandas. To harmonize artifact detection and phasic decomposition, EDA signals were downsampled to 8 Hz. Motion artifacts were identified using the MIT Media Lab EDA-Explorer pipeline and interpolated where detected. Cleaned signals were low-pass filtered and decomposed into tonic and phasic components using cvxEDA as implemented in NeuroKit2. Event-related analyses were aligned to stimulus onset triggers. Because SCRs have a delayed latency, responses within the first 4s were treated as orienting responses, while later SCRs were interpreted as stimulus-related chill responses. The results provide evidence that spoken language can reliably elicit physiological chill responses. Speech stimuli produced significant SCR peak responses, supporting the hypothesis that non-melodic language can move listeners physiologically. However, physiological and subjective indices dissociated: although SCR peaks occurred frequently, participants did not consistently report these experiences as chills. Comparisons between speech and music showed that the occurrence of physiological chills did not differ reliably between the two stimulus types. However, when physiological chills occurred, music elicited stronger responses than speech, as reflected in higher SCR peak counts. At the same time, substantial interindividual variability was observed, consistent with previous evidence that only a subset of listeners reliably experiences chills. These findings suggest that spoken language can elicit bodily responses comparable in occurrence, though not intensity, to those elicited during music listening. Importantly, the dissociation between physiological and subjective measures indicates that language may affect listeners at a physiological level even when they do not consciously label the experience as a chill. This has implications for the study of affective speech perception, aesthetic language experience, and the shared neurobiological mechanisms underlying music and language. The results support the view that aesthetic responses to language are not limited to semantic or literary meaning but may also emerge from the perceptual and affective qualities of speech sound itself.